EPSR-test-ex32 mixato: file selection and error paths

Program(s): mixato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises the interactive file-selection dialogue shared by the EPSR setup tools [1] (`search_for_files`) inside `mixato`, with a real directory listing, plus the error branches for a non-existent file and a missing template. These paths matter because every box-building program of EPSR [2] selects its input files through them.

Code exercised

Fortran (original, fortran_src/):

- `mixato.f`: `mixato` (number of files from the command line, template check, output name with extension)
- `file_and_directory_commands.f`: `search_for_files` (answers `u`, `y`, `n`, `e`; rewind/wrap-around), `get_full_filename`
- `system_commands.f`: reading `system_commands.txt` (`system_ls`, `system_termination`)
- `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`)

Rust (port, rust/epsr-core/src/):

- `tools/mixato.rs`: `run`
- `tools/setup_common.rs`: `search_for_files`, `SysCmds`, `get_full_filename`
- `ato.rs`: `minitc`, `moloutc`, `ftype`

Physical system

A 12-molecule box of 5 water and 7 dimethyl sulphoxide molecules (85 atoms) at 0.08 atoms/Å³ and 300 K, cubic side 10.204 Å (`mix2.ato`); the molecules are those of the EPSR-test DMSO-water system [3]. The water file is `Water.ato`, a renamed copy of `WaterMolecule.ato`, so its stored template does not exist.

Inputs

`DMSO.ato`, `Water.ato` (= EPSR-test `WaterMolecule.ato` renamed), a Linux `system_commands.txt` (`system_ls ls`, `system_termination /`) and `mixato.stdin`; command `mixato 2`. Answers: `u` (go back) on the first entry, which re-positions at the end of the list, `y` accept, `n` rejections with wrap-around, `e` (leave the list) and typing `NoSuchFile` ("Problems with .ato file ... Try again"), then output name `mix2.out` (stored as `mix2.ato`).

How the accuracy was verified

`commands.txt` (1 program call, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

The listing is real: the reference `dirlist.txt` contains `./DMSO.ato` and `./Water.ato` (from `ls ./*.ato`); the answers were chosen so that the result does not depend on the locale sort order of `ls`. The missing-template branch is taken: the water molecule type is written with template `Water` (the file name) in `mix2.ato`. Both output files are identical.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).

[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.